

FSD Laboratory 04

Aim: Write server-side script in PHP to perform form validation and create database application using PHP and MySQL to perform insert, update, delete and search operations.

Objectives:

1. To understand Server-side Scripting.
2. To learn database connectivity using PHP-MySQL.
3. To perform insert, update, delete and search operations on database.

Theory:

1. PHP Architecture.

PHP (Hypertext Preprocessor) follows a **server-side scripting architecture** where the PHP code runs on the web server and generates HTML that is sent to the client's browser.

Components of PHP Architecture

2. **Client (Browser)**
 - a. The user sends a request through a web browser such as Chrome or Firefox.
 - b. The request is sent to the web server for processing.
3. **Web Server**
 - a. The web server (like Apache or Nginx) receives the request.
 - b. It identifies the requested PHP file and passes it to the PHP interpreter.
4. **PHP Interpreter / PHP Engine**
 - a. The PHP engine processes the PHP script.
 - b. It executes the server-side code and performs operations such as database access and file handling.
5. **Database**
 - a. PHP interacts with databases such as MySQL, MariaDB, or PostgreSQL.
 - b. It retrieves or stores data as required by the application.
6. **Response to Client**
 - a. After processing, the PHP engine generates HTML output.
 - b. The web server sends the generated HTML page back to the user's browser.

2. Steps for Database connectivity in PHP.

Steps for Database Connectivity in PHP

Database connectivity in PHP is used to connect a PHP application with a database such as MySQL to store and retrieve data.

1. Create a Database

First, create a database in MySQL using phpMyAdmin or SQL command.

Example:

CREATE DATABASE student_db;

2. Create a Table

Create a table inside the database to store data.

Example:

```
CREATE TABLE students(  
id INT PRIMARY KEY,  
name VARCHAR(50),  
email VARCHAR(50)  
);
```

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6. Close Connection

Close the database connection after completing operations.

Example:

FAQ:

1. What are the advantages of Server-side Scripting?

Server-side scripting refers to scripts that run on the **web server** instead of the user's browser. Languages such as PHP, Python, ASP.NET, and Node.js are used for server-side scripting.

1. Better Security

Server-side scripts run on the server, so the source code is hidden from users. This helps protect sensitive information like database queries and business logic.

2. Database Connectivity

Server-side scripting easily connects with databases such as MySQL, PostgreSQL, and Oracle to store, retrieve, update, and delete data.

3. Dynamic Web Pages

It allows developers to create dynamic web pages that change based on user input, login information, or database data.

4. Centralized Processing

All processing happens on the server, making it easier to manage updates, security, and application logic.

5. Supports Complex Applications

Server-side scripting is suitable for developing complex web applications such as e-commerce websites, content management systems, and student portals.

2. What is XAMPP and phpMyAdmin?

XAMPP

XAMPP is a **free and open-source web server package** used to run web applications locally on a computer. It allows developers to test PHP and MySQL applications without using an online server.

XAMPP stands for:

- **X** – Cross Platform
- **A** – Apache Web Server
- **M** – MySQL / MariaDB Database
- **P** – PHP
- **P** – Perl

phpMyAdmin

phpMyAdmin is a **web-based tool used to manage MySQL or MariaDB databases** through a graphical user interface.

Functions of phpMyAdmin:

- Create and delete databases
- Create, edit, and delete tables
- Insert, update, and delete records
- Run SQL queries
- Import and export databases

3. What are the two ways to connect to a database in PHP?

Two Ways to Connect to a Database in PHP

In PHP, there are two main methods used to connect a web application to a MySQL database.

1. MySQLi (MySQL Improved)

MySQLi is an improved extension used to connect PHP with MySQL databases. It supports both **procedural and object-oriented programming** styles.

Example:

```
$conn = mysqli_connect("localhost","root","","student_db");

if(!$conn){
    die("Connection failed: " . mysqli_connect_error());
}
```

Features:

- Supports prepared statements
- Faster and more secure than old MySQL extension
- Supports procedural and object-oriented style

2. PDO (PHP Data Objects)

PDO is a database access method that allows PHP to connect with **multiple types of databases** such as MySQL, PostgreSQL, SQLite, and Oracle.

Example:

```
$conn = new PDO("mysql:host=localhost;dbname=student_db","root","");
```

Output: Screenshots of the output to be attached.

Sample Problem Statements:

PHP CRUD Operations

1. Student can create a PHP form or use existing/ implemented HTML form for Student's Registration System with the fields mentioned: First name, Last name, Roll No/ID, Password, Confirm Password, Contact number and perform following operations

1. Insert student details - First name, Last name, Roll No/ID, Password, Confirm Password, Contact number
2. Delete the Student records based on Roll no/ID
3. Update the Student details based on Roll no/ID - Example students can update their contact details based on searching the record with Roll no.
4. Display the Updated student details or View the Students record in tabular format.

Apply Form Validation on the necessary fields using PHP/Javascript

2. Student can create a PHP form or use existing/ implemented HTML form for Library Management System with the fields mentioned: Book name, ISBN No, Book title, Author name, Publisher name and perform following operations

1. Insert Book details - Book name, ISBN No, Book title, Author name, Publisher name
2. Delete the Book records based on ISBN No
3. Update the Book details based on ISBN No - Example students can update wrong entered book details based on searching the record with ISBN No.
4. Display the Updated Book details or View the Book Details records in tabular format.

Apply Form Validation on the necessary fields using PHP/Javascript

3. Student can create a PHP form or use existing/ implemented HTML form for Employee Management System with the fields mentioned: Employee name, Employee ID, Department_name, Phone number, Joining Date and perform following operations

1. Insert Employee details - Employee name, Employee ID, Department_name, Phone number, Joining Date
2. Delete the Employee records based on Employee ID
3. Update the Employee details based on Employee ID - Example students can update Employee details based on searching the record with Employee ID.
4. Display the Updated Employee details or View the Employee Details records in tabular format.

Apply Form Validation on the necessary fields using PHP/Javascript

4. Student can create a PHP form or use existing/ implemented HTML form for Flight Booking Management System with the fields mentioned: Passenger name, From, to, date, Departure date, Arrival date, Phone number, Email ID and perform following operations

1. Insert Passenger details - Passenger name, From, to, date, Departure date, Arrival date, Phone number, Email ID
2. Delete the Passenger records based on Phone Number
3. Update the Passenger details based on Phone Number - Example students can update Flight Booking details based on searching the record with Phone Number.
4. Display the Updated Flight Booking details or View the Flight Booking Details records in tabular format.

Apply Form Validation on the necessary fields using PHP/Javascript.

Technologies Student Should Use:

XAMPP

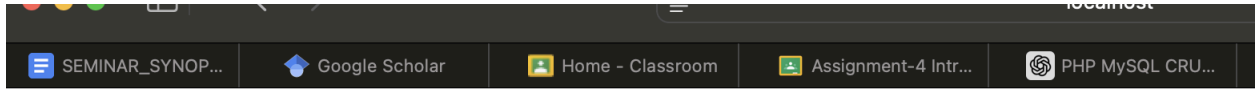
PHP for Server-side Scripting

MySQL as a backend Database



Dr. Vishwanath Karad
MIT WORLD PEACE
UNIVERSITY | PUNE
TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

School of Computer Engineering & Technology
Class: Third Year B.Tech CSE (Semester V)
Course: Full Stack Development



Student Records

Roll No	First Name	Last Name	Contact	Delete	Update
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Student Registration

First Name: Arhaan

Last Name: Mulani

Roll No: 42



Password:

Confirm Password:

Contact Number: 4344344

Register

[View Students](#)



Student Records

Roll No	First Name	Last Name	Contact	Delete	Update
42	Arhaan	Mulani	4344344	Delete	Update
56	ggg	grg	4344343rr4	Delete	Update

```
create mode 100644 student_crud/db.php
create mode 100644 student_crud/delete.php
create mode 100644 student_crud/index.php
create mode 100644 student_crud/insert.php
create mode 100644 student_crud/style.css
create mode 100644 student_crud/update.php
create mode 100644 student_crud/view.php
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git push origin assignment4
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 14 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (10/10), 2.03 KiB | 2.03 MiB/s, done.
Total 10 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'assignment4' on GitHub by visiting:
remote:   https://github.com/arhaanmulani/arhaan_FSDL/pull/new/assignment4
remote:
To https://github.com/arhaanmulani/arhaan_FSDL.git
 * [new branch]      assignment4 -> assignment4
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git checkout assignment4
Already on 'assignment4'
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git rm file1.txt file2.txt file3.txt LICENSE
rm 'LICENSE'
rm 'file1.txt'
rm 'file2.txt'
rm 'file3.txt'
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git commit -m "Assignment 4: Removed unnecessary files from branch"
[assignment4 453dc77] Assignment 4: Removed unnecessary files from branch
4 files changed, 204 deletions(-)
delete mode 100644 LICENSE
delete mode 100644 file1.txt
delete mode 100644 file2.txt
delete mode 100644 file3.txt
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git push origin assignment4
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 14 threads
Compressing objects: 100% (1/1), done.
Writing objects: 100% (2/2), 267 bytes | 267.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/arhaanmulani/arhaan_FSDL.git
 05eb9cb..453dc77 assignment4 -> assignment4
arhaan@Arhaans-MacBook-Pro arhaan_FSDL %
```

```
Restarting XAMPP for Mac OS X 8.2.4-0...
XAMPP: Stopping Apache...ok.
XAMPP: Stopping MySQL...not running.
XAMPP: Stopping ProFTPD...ok.
XAMPP: Starting Apache...ok.
XAMPP: Starting MySQL...ok.
XAMPP: Starting ProFTPD...ok.
arhaan@Arhaans-MacBook-Pro ~ % sudo /Applications/XAMPP/xamppfiles/xampp startmysql
Password:
XAMPP: Starting MySQL...ok.
arhaan@Arhaans-MacBook-Pro ~ % cd /Applications/XAMPP/htdocs/student_crud
arhaan@Arhaans-MacBook-Pro student_crud % git init
Initialized empty Git repository in /Applications/XAMPP/xamppfiles/htdocs/student_crud/.git/
arhaan@Arhaans-MacBook-Pro student_crud % cd ~/Documents/arhaan_FSDL
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git checkout main
git pull origin main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
From https://github.com/arhaanmulani/arhaan_FSDL
* branch      main      -> FETCH_HEAD
Already up to date.
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git checkout -b assignment4
Switched to a new branch 'assignment4'
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git branch
  assignment2
  assignment3
* assignment4
  main
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % mv "/Applications/XAMPP/htdocs/student_crud" .
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % ls
file1.txt  file2.txt  file3.txt  LICENSE  student_crud
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git add student_crud
error: 'student_crud/' does not have a commit checked out
fatal: adding files failed
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % cd student_crud
arhaan@Arhaans-MacBook-Pro student_crud % rm -rf .git
arhaan@Arhaans-MacBook-Pro student_crud % cd ..
git add student_crud
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git commit -m "Assignment 4: Added student CRUD PHP project"
[assignment4 05eb9cb] Assignment 4: Added student CRUD PHP project
 7 files changed, 184 insertions(+)
 create mode 100644 student_crud/db.php
 create mode 100644 student_crud/delete.php
```